

Tom de Groot

Chemistry - Math - AI

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TECHNICAL STACK

AI/Evaluation: PyTorch, TensorFlow, agentic workflows, RL environments, benchmark design

Data: pandas, NumPy, SQL, RDKit, scientific data quality, provenance, versioning

Formal: Lean 4, mathlib, Lake, CI/CD for proofs

Tools: Git, AWS, Docker, Linux/HPC, APIs

Languages: Python, Lean, JavaScript
Spoken: Dutch, English

EDUCATION

BSc Chemistry

University of Groningen
2021 - 2025
Track: Chemistry of Life

Thesis: Evaluating and Optimizing NMR Prediction with Graph Neural Networks

AWARDS

Chess.com Member of the Year 2021

Harmonic Rising Mathematician Award

ABOUT ME

AI engineer at **SoundImports** and BSc Chemistry graduate from the University of Groningen. My thesis in the Pollice Group focused on evaluating and optimizing graph neural networks for NMR prediction, including molecular-data cleaning, HPC training, and controlled model comparison.

I build scientific-AI and agentic systems around explicit data lineage, reproducible environments, hard evaluations, and failure-driven iteration - from production prediction systems to deterministic simulators and Lean-verified mathematics.

SCIENTIFIC AI, EVALUATION & DATA

Bazaar_ML - Simulator, Certification & RL Environment

2026 - Present

Built a simulator-first ML stack that ingests versioned public source data, compiles heterogeneous mechanics into executable records, simulates complete trajectories, emits deterministic traces, and exposes agent-training hooks.

Designed schema-frozen certification and provenance. The committed matrix contains **49,248 cases**, **99.85%** executable sampling-pool coverage, explicit parser/content hashes and unsupported primitives, and zero recorded execution failures.

NMR Prediction with Graph Neural Networks

2024 - 2025

Curated molecular and NMR data and ran controlled GNN experiments on HPC. Compared architectures and outcomes through a hypothesis-test-measure-iterate workflow, with attention to identifiers, scientific context, split design, and evaluation quality.

RESEARCH & FORMALIZATION

FirstProof Problem 9 - Lean-Verified Agent Workflow

Feb 2026

Orchestrated multiple LLM agents to find an informal solution and formalize it in Lean 4. Lean compilation provided a hard evaluator for every accepted step; the pinned workspace contains the only solved problem in that repository. Repository

Prime Number Theorem And (PNT+)

Late 2025 - Present

Co-authored multiple merged Lean proofs in the open-source PNT+ project and use compilation, tests, and adversarial review to evaluate model-generated mathematics.
Merged PRs

ENGINEERING PROJECTS

BaanSoliciteren - Agentic Job Automation

2025 - 2026

Built a Python/Gmail MCP pipeline to ingest vacancy pages and email, normalize them into JSON, score fit, track state, and archive outcomes with an auditable workflow.
GitHub

PROFESSIONAL EXPERIENCE

AI Engineer - SoundImports

Present

Develop prediction algorithms and AI automations; deploy containerized services on AWS and build full-stack applications across data, model, API, and production layers.

Teamspot - Regional Director / QA

2022 - 2024

Coordinated regional school projects, trained users, and bug-tested web, app, API, and integration workflows in Jira.

Transcom (for Odido)

2024

Maintained consistently strong customer-satisfaction results and learned to explain technical and operational issues clearly.

Chess.com

2022

Moderated and proctored international chess events; recognized as Member of the Year 2021 for community ownership and coaching.